Experiment 7

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Title: Temporal Difference Learning using TD(0) Algorithm

Software Required:

- Python
- Gymnasium

AIM

To implement the TD(0) Temporal Difference Learning algorithm for estimating state values.

THEORY

Temporal Difference Learning updates the value of a state using the immediate reward and the estimated value of the next state. TD(0) learns after each transition without waiting for the complete episode.

PROCEDURE

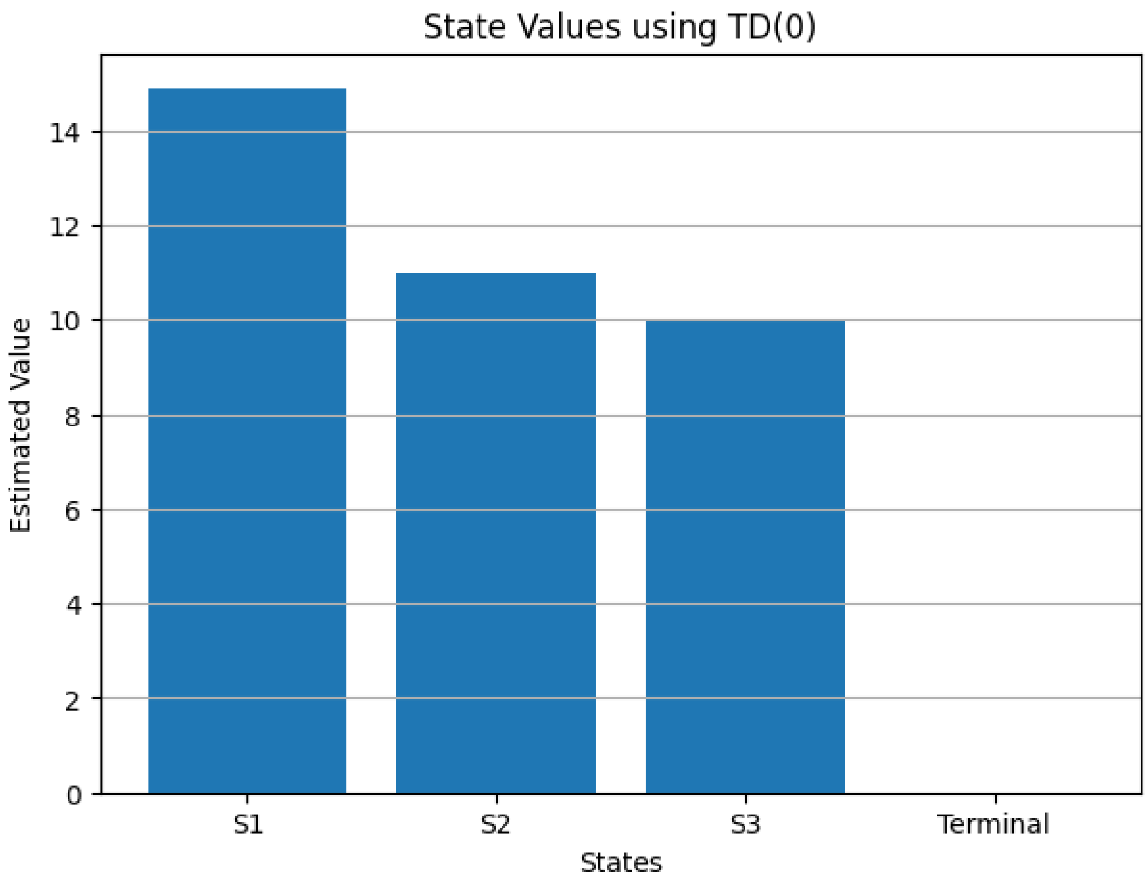
1. Open the Python environment.
2. Import Gymnasium and required libraries.
3. Create the Reinforcement Learning environment.
4. Initialize the state-value function.
5. Set the learning rate and discount factor.
6. Reset the environment to obtain the initial state.
7. Select an action and observe the next state and reward.
8. Calculate the TD target and TD error.
9. Update the value of the current state using the TD(0) rule.
10. Repeat the process for multiple episodes and observe the learned values.

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

# States
states = ['S1', 'S2', 'S3', 'Terminal']

# Rewards
rewards = [5, 2, 10, 0]

# Parameters
alpha = 0.1
gamma = 0.9
episodes = 100

# Initialize Value Function
V = np.zeros(len(states))

# TD(0) Learning
for episode in range(episodes):
    for s in range(len(states)-1):
        td_target = rewards[s] + gamma * V[s+1]
        td_error = td_target - V[s]
        V[s] = V[s] + alpha * td_error

# Output
print("Estimated State Values using TD(0):\n")
for i in range(len(states)):
    print(f"{states[i]} : {V[i]:.2f}")

# Graph
plt.figure(figsize=(7,5))
plt.bar(states, V)

plt.title("State Values using TD(0)")
plt.xlabel("States")
plt.ylabel("Estimated Value")
plt.grid(axis='y')

plt.show()
```

OUTPUT:

```
Estimated State Values using TD(0):

S1 : 14.88
S2 : 11.00
S3 : 10.00
Terminal : 0.00
```



RESULT

The TD(0) algorithm was successfully implemented and the state values were estimated using Temporal Difference Learning.